

About Multi-Stage Structures

In most cases, each outlet structure is treated independently, but you can put the structures in series, that is, create a multi-stage structure. This enables you to direct the outflows from culvert B, C and perforated riser, and weirs A, B, C and D through culvert A. The structure with the least capacity at any given stage controls the final outflow.

A typical multi-stage structure consists of a standpipe weir (box or circular) with an orifice attached to the lower side of the standpipe, near the invert. The standpipe is connected to the upstream end of culvert A. All flows from the orifice and weir are combined and directed to and through culvert A.

NOTE:

Culvert A does not have the Multi-Stage option. It is always the final outflow structure.

You can have more than one orifice or weir. You can specify orifices, weirs B, C, and D, each at different elevations. Select the multi-stage option when entering data.

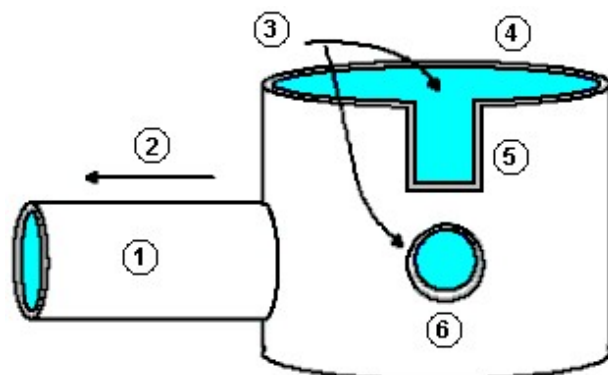
IMPORTANT:

You must set multi-stage orifice elevations at least 0.01 feet above ground to prevent culvert A from taking control too early in the process.

Best Practice

When using the Multi-stage option, Hydraflow Hydrographs Extension checks the head elevation produced by culvert A. This value is then used as a tailwater elevation against other multi-stage structures. As the head increases, the outflows from the orifices and weirs decrease. When the head equals the current stage, culvert A is the controlling structure and the contributing flows from the orifices and weirs diminish.

The following is a typical multi-stage structure.



(1) Culvert A	(3) Inflow	(5) Weir B
(2) Outflow	(4) Weir A - Riser	(6) Culvert B

